EX1 - sol

Saturday, October 31, 2026 10:07 AM

1) (a) $\mathcal{F}(f*g) = \int_{x=-\infty}^{\infty} f*g(x)\cdot e^{-2\pi i ux}\,dx = \int_{x=-\infty}^{\infty}\int_{\alpha=-\infty}^{\infty} f(\alpha)\,g(x-\alpha)\,d\alpha \cdot e^{-2\pi i ux}\,dx =$

$= \int_{\alpha} f(\alpha)\left(\int_{x} g(x-\alpha)e^{-2\pi i ux}\,dx\right)d\alpha \overset{\substack{t=x-\alpha \\ x=t+\alpha}}{=} \int_{\alpha} f(\alpha)\left(\int_{t} g(t)e^{-2\pi i u(t+\alpha)}\,dt\right)d\alpha =$

$= \int_{\alpha} f(\alpha)e^{-2\pi i u\alpha}\cdot\left(\int_{t} g(t)e^{-2\pi i ut}\,dt\right)d\alpha = \int_{\alpha=-\infty}^{\infty} f(\alpha)e^{-2\pi i u\alpha}\,d\alpha \cdot \int_{t=-\infty}^{\infty} g(t)e^{-2\pi i ut}\,dt =$

$= F\cdot G$

(b) $f*g = \int_{\alpha=-\infty}^{\infty} f(\alpha)\,g(x-\alpha)\,d\alpha \overset{\substack{t=x-\alpha \\ \alpha=x-t}}{=} \int_{t=-\infty}^{\infty} f(x-t)\,g(t)\,dt = \int_{t=-\infty}^{\infty} g(t)f(x-t)\,dt = g*f$

(c) $f*(\alpha g+\beta h) = \int_{\lambda=-\infty}^{\infty} f(\lambda)\,(\alpha g+\beta h)(x-\lambda)\,d\lambda = \int_{\lambda=-\infty}^{\infty} f(\lambda)\,(\alpha g(x-\lambda)+\beta h(x-\lambda))\,d\lambda =$

$= \int_{\lambda=-\infty}^{\infty} f(\lambda)\cdot\alpha g(x-\lambda)+f(\lambda)\cdot\beta h(x-\lambda)\,d\lambda = \alpha\int_{\lambda=-\infty}^{\infty} f(\lambda)g(x-\lambda)\,d\lambda + \beta\int_{\mu=-\infty}^{\infty} f(\mu)\cdot h(x-\mu)\,d\mu =$

$= \alpha(f*g)+\beta(f*h)$

(d) $f(x)*g(x-d) = \int_{\alpha=-\infty}^{\infty} f(\alpha)\cdot g(x-d-\alpha)\,d\alpha = (f*g)(x-d)$

2) lets look at it in fourier domain -

$\mathcal{F}\left(\frac{d}{dx}\{f(x)*g(x)\}\right) \overset{\text{fourier derivation}}{=} 2\pi i u\,\mathcal{F}(f(x)*g(x)) \overset{\text{convolution theorem}}{=} 2\pi i u F(u)G(u) =$

$= \mathcal{F}\left(\frac{d}{dx}f(x)\right)\cdot\mathcal{F}(g(x)) = \mathcal{F}\left(\left(\frac{d}{dx}f(x)\right)*g(x)\right)$

(fourier derivation; convolution theorem)

taking $\mathcal{F}^{-1}$ on both sides we get

$\frac{d}{dx}\{f(x)*g(x)\} = \left(\frac{d}{dx}f(x)\right)*g(x)$

3) (a) let f be a symmetric function and $\mathcal{F}(f) = F$

$\overset{t=-x}{=} \int_{-\infty}^{\infty} e^{-2\pi i(-u)x}\cdot\int_{-\infty}^{\infty} e^{-2\pi i ux}$

3) (a) let f be a symmetric function and $\mathcal{F}(f) = F$

$$\mathcal{F}(f(x))(-u) = F(-u) = \int_{x=-\infty}^{\infty} f(x) e^{-2\pi i(-u)x}\, dx \overset{t=-x}{=} \int_{t=-\infty}^{\infty} f(-t) e^{-2\pi i u x}\, dx = \int_{t=-\infty}^{\infty} f(t) e^{-2\pi i u x}\, dx = F(u)$$

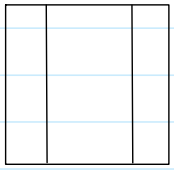
(b) let f be a symmetric and real function and $\mathcal{F}(f) = F$

$$F(u) = \int_{x=-\infty}^{\infty} f(x) e^{-2\pi i u x}\, dx = \int_{x=-\infty}^{\infty} f(x)\left(\cos(-2\pi u x) + i \sin(-2\pi u x)\right) dx =$$

$$= \int_{x=-\infty}^{\infty} f(x) \cdot \cos(-2\pi u x)\, dx + i \underbrace{\int_{x=-\infty}^{\infty} f(x) \sin(-2\pi u x)\, dx}_{0}$$

Since sin is odd and $f(x)$ is even their product $f(x)\sin(-2\pi u x)$ is odd integrating it from $-\infty$ to ∞ cancel out.
Therfore, the imaginary part of $F(u)$ is always 0, so $F(u)$ is a real.
From 3a, we know it is symmetric

4) (a)

$f =$ 

two liner (1-pixel width)

each dot acts as a delta function as explained in class, convoluting with each delta shifts and duplicates

(b) let $l(x,y)$ be the line, and $d(x,y)$ be the dots
by convolution theorem -

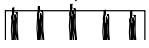
$$F(u,v) = \mathcal{F}(f(x,y)) = \mathcal{F}(l(x,y) * d(x,y)) = \mathcal{F}(l(x,y)) \cdot \mathcal{F}(d(x,y)) = L(u,v) \cdot D(u,v)$$

so

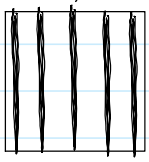
$$|F(u,v)| = |L(u,v)| \cdot |D(u,v)|$$

let's look at $|L(u,v)|$ - since it's a line that stretch the entire image vertically its fourier is a horrizontal and centered line $|L(u,v)| =$ 

let's look at $D(u,v)$
$d(x,y)$ is an addition of two dots, by linearity, their $D(u,v)$ is an addition of fourier of two shifted dots.
a single shifted dot have a constant magnitude of for all (u,v) with some phase shift depending on the shift.
adding two of these, with a different shift, will cause a cosine wave pattern between 0 and 2, with its direction depending on the direction between the dots.



Pattern between 0 and 2, with its direction depending on the direction between the dots.

$|D(u,v)| =$ 

so $|F(u,v)|$ will be a horizontal line fading in a cosine pattern

$|F(u,v)| =$ 

5) $$G(u,v) = \begin{cases} F(u,v) & \text{if } 0 \le u < N \text{ and } 0 \le v < M \\ 0 & \text{otherwise} \end{cases}$$

$$f(x,y) = \sum_{u=0}^{N-1} \sum_{v=0}^{M-1} F(u,v) e^{2\pi i \left(\frac{ux}{N} + \frac{vy}{M}\right)}$$

$$g(x,y) = \sum_{u=0}^{2N-1} \sum_{v=0}^{2M-1} G(u,v) e^{2\pi i \left(\frac{ux}{2N} + \frac{vy}{2M}\right)} = \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} F(u,v) e^{2\pi i \left(\frac{ux}{2N} + \frac{vy}{2M}\right)}$$

↑ by G def

so $g(x,y)$ is identical to $f(x,y)$, aside from a $\frac{1}{2}$ scaling on $\frac{ux}{N}$ and $\frac{vy}{M}$

notice $g(2x, 2y) = f(x,y)$

so the image is stretched, with the even coordinates having the same value as the original image

6) $$G(u,v) = \begin{cases} F\left(\frac{u}{2}, \frac{v}{2}\right) & \text{if } u \text{ is even and } v \text{ is even} \\ 0 & \text{otherwise} \end{cases}$$

$$g(x,y) = \sum_{u=0}^{2N-1} \sum_{v=0}^{2M-1} G(u,v) e^{2\pi i \left(\frac{ux}{2N} + \frac{vy}{2M}\right)}$$

let's do a coord change $u = 2p$, $v = 2q$ for even u,v

$$= \sum_{p=0}^{N-1} \sum_{q=0}^{M-1} G(2p, 2q) e^{2\pi i \left(\frac{2px}{2N} + \frac{2qy}{2M}\right)} + \sum_{\text{odd } u,v} 0 = \sum_{p=0}^{N-1} \sum_{q=0}^{M-1} F(p,q) e^{2\pi i \left(\frac{px}{N} + \frac{qy}{M}\right)} = f(x,y)$$

due to the periodic nature of the discrete fourier transform, the image is repeated for $x > N$ $y > M$

due to the periodic nature of the discrete fourier transform, the image is repeated for $x \geq N, y \geq M$.

$$g(x,y) = f(x \bmod N, y \bmod M)$$

7) (a) according to the sampling theorem, as showed in class, if $F(u)$ of some $f(x)$ is band limited $- \exists \Omega_0$ s.t. $\forall |u| > \Omega_0, F(u) = 0$
then we can sample it discretly with $\Delta u > 2\Omega_0 \Rightarrow \Delta x < \frac{1}{2\Omega_0}$, and reconstruct it.

$f_1(x) = \sin(\alpha x)$ so $F_1(u) = \frac{1}{2i}\delta(u - \frac{\alpha}{2\pi}) - \frac{1}{2i}\delta(u + \frac{\alpha}{2\pi})$ from lecture handouts

notice - $\delta(u - \frac{\alpha}{2\pi}) \neq 0 \iff u = \frac{\alpha}{2\pi}$ and $\delta(u + \frac{\alpha}{2\pi}) \neq 0 \iff u = -\frac{\alpha}{2\pi}$

lets take $\Omega_0 > |\frac{\alpha}{2\pi}|$, now $\forall |u| > \Omega_0$ $\delta(u - \frac{\alpha}{2\pi})$ and $\delta(u + \frac{\alpha}{2\pi})$ will never be non zero, so $F_1(u) = 0$
this means $f_1(u)$ is band limited $\Rightarrow$ the minimal allowed distance between samples is

assuming $\alpha > 0$

$$2\frac{1}{\Omega_0} = \frac{1}{\frac{|\alpha|}{2\pi}} \overset{\alpha > 0}{=} \frac{2\pi}{\alpha}$$

(b) $f_2(x) = \begin{cases} 1 & -1 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$ so $F_2(u) = 2\,\mathrm{sinc}(2\pi u) = \frac{2\sin(2\pi u)}{2\pi u} = \frac{\sin(2\pi u)}{\pi u}$ from handouts

$f_2(u)$ is not band limited, take $u^* = n + \frac{1}{4}$ $\forall n \in \mathbb{N}$

$$F_2(u^*) = \frac{\sin(2\pi \cdot (n + \frac{1}{4}))}{\pi \cdot (n + \frac{1}{4})} = \frac{\sin(2\pi n + \frac{\pi}{2})}{\pi n + \frac{\pi}{4}} = \frac{1}{\pi n + \frac{\pi}{4}} > 0$$

so there is no Ω_0 where above it $F_2(u)$ is 0 $\Rightarrow$ $f_2(u)$ is not bandlimited and cannot be sampled discretly without losing information

(c) $f_3(x) = f_1(x) * f_2(x)$ by convolution theorem

$$F_3(u) = F_1(u) \cdot F_2(u) = \left(\frac{1}{2i}\delta(u - \frac{\alpha}{2\pi}) - \frac{1}{2i}\delta(u + \frac{\alpha}{2\pi})\right) \cdot \frac{\sin(2\pi u)}{\pi u} =$$

$$= \frac{\sin(2\pi u)}{2i\pi u}\delta(u - \frac{\alpha}{2\pi}) - \frac{1}{2i\pi u}\delta(u + \frac{\alpha}{2\pi})$$

again, we can take $\Omega_0 > |\frac{\alpha}{2\pi}|$ and both δ functions will be 0 $\forall |u| > \Omega_0$
so $F_3(u)$ is band limited and the minimal sampling distance is $\frac{2\pi}{\alpha}$, as was in (a)

8) (a) The final mix resembles the image from which the magnitude was taken much more than the phase
The difference between Mix_1 and mag_2 as well as mix_2 and mag_1, is the position of the stripes, their size and spacing and orientation stay the same - controlled by magnitude but phase controls their position

The difference between mix_1 and mix_2 is the position of the stripes, their size and spacing and orientation stay the same - controlled by magnitude, but phase controls their position.

(d) for these more natural images, the final mix mostly resembles the phase. The case here is reversed because features such as edge and shapes have to be in the correct spot for us to recognize them, where in Im_1 and Im_2 the image is dominated by the frequency itself, not its location.

3) $g(x) = f(sx) \qquad s>0$

$$G(u) = \int_{x=-\infty}^{\infty} g(x) e^{-2\pi i u x}\,dx = \int_{x=-\infty}^{\infty} f(sx) e^{-2\pi i u x}\,dx \overset{\substack{t=sx \\ x=\frac{t}{s}}}{=} \int_{t=-\infty}^{\infty} f(t) e^{-2\pi i u \frac{t}{s}}\,d\frac{t}{s} =$$

$$= \frac{1}{s}\int_{t=-\infty}^{\infty} f(t) e^{-2\pi i \frac{u}{s}\cdot t}\,dt = \frac{1}{s}\cdot F\left(\frac{u}{s}\right)$$